

Polydimethylsiloxane-based Antifouling Polymers with Tunable Self-healing Properties in Aqueous Environments

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Electronic Supplementary Information

Abstract Development of polymers with underwater self-healing and antifouling properties is crucial, particularly in harsh marine environments. In this study, polydimethylsiloxane (PDMS)-based antifouling polymers with tunable self-healing capabilities in aqueous conditions were fabricated by incorporating amphiphilic segments and Fe³⁺-catechol dynamic coordination crosslinking. The microphase formed within the PDMS matrix imparted static antifouling properties to the coatings. The mechanical properties of the damaged sample were restored at room temperature in an aqueous environment for 24 h, achieving a self-healing efficiency of almost 100%. The synthesized material exploited the dynamic coordination between Fe³⁺ and catechol to facilitate underwater self-healing. No bacterial adhesion was observed at the scratch site after the coating was repaired. This material enables the long-term antifouling and autonomous repair of marine vessels and sensors, thereby reducing maintenance costs.

Keywords Antifouling polymers; Self-healing; Polydimethylsiloxane (PDMS)-based polymers

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INTRODUCTION

Antifouling materials are essential for devices used in marine environments because they effectively reduce biofouling adhesion, friction resistance, fuel consumption, and economic cost upon navigation.^[1–3] Commonly used resins for antifouling coatings include polydimethylsiloxane (PDMS),^[4,5] epoxy resins,^[6] and acrylic resins.^[7] PDMS exhibits low surface energy and low modulus, making it highly effective in repelling fouling.^[8–10] To improve the static antifouling ability of PDMS coatings, antifouling agents,^[11,12] quaternary ammonium salts,^[13] and hydrophilic components^[14,15] have been incorporated into the polymer coating matrix. Nonetheless, PDMS-based antifouling coatings are fragile and prone to damage, resulting in loss of their antifouling function in aqueous environments.^[16] The deposition and accumulation of marine organisms on the damaged surface accelerates microbiologically influenced corrosion.^[17–20]

Recently developed self-healing strategies that can heal damage without human intervention, especially in harsh ma-

rine environments, lead to higher polymer stability and a longer life span.^[21–25] Intrinsic methods, such as incorporating supramolecular interactions or dynamic covalent bonding, and extrinsic methods such as embedding microcapsules or nanocontainers within a matrix, have been employed to construct self-healing materials.^[24,26–31] Multi-strength hydrogen bonding interactions have been utilized to construct PDMS composites that can self-heal in air.^[32–34] Self-healing in an aqueous environment without human intervention is especially essential for PDMS-based antifouling coatings.^[35] Hydrogen bonds,^[36] boronic-ester bonds,^[37] metal-catechol interactions,^[38–41] and imine bonds^[42] have been utilized to achieve underwater self-healing. Among them, Fe³⁺-catechol complexation exhibits dynamic behavior in the presence of water, making it highly suitable for applications in humid environments, including seawater. However, strategies for developing antifouling polymers with self-healing properties in aqueous environments remain limited.

In this study, PDMS-based antifouling polymers with tunable self-healing properties that function in aqueous environments were constructed by incorporating Fe³⁺-catechol crosslinking. The coating exhibited excellent antifouling properties owing to the microphase-separated morphology formed by amphiphilic segments. When damaged, the mechanical properties of the initial polymer materials can be

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completely restored after immersion in artificial seawater or deionized water. Self-healing behavior can be tuned by adjusting the content of the dynamic Fe^{3+} -catechol crosslink points. Compared to traditional antifouling coatings, PDMS-based polymers exhibit efficient antifouling and tunable self-healing properties in aqueous environments. This material enables the long-term antifouling and autonomous repair of marine vessels and sensors, thereby reducing maintenance costs.

EXPERIMENTAL

Materials

α,ω -Aminopropyl-terminated poly(dimethyl siloxane) (APT-PDMS) ($M_n=3000$ g/mol) was purchased from Hangzhou Silon Materials. Tech. Co. Ltd. Superdry tetrahydrofuran (THF, 99.9%) and *N,N*-dimethyl formamide (DMF, 99.9%) were obtained from J&K Scientific. Super-dry TEA (99.5%) was obtained from Merck. 2,2'-Azobis(2-methylpropionitrile) (AIBN) was obtained from Aladdin and recrystallized from methanol. Isophorone diisocyanate IPDI (99%) was obtained from Aladdin. Dodecafluoroheptyl methacrylate (DFMA) was purchased from Harbin Xuejia Fluorine Silicon Chemical Co. Ltd. *N*-Hydroxysuccinimide (NHS, 98%), Boc-Lys(Boc)-OH (BL, 98%), dopamine hydrochloride (DA-HCl, 98%), and 1-ethyl-3-(3-dimethylaminopropyl)-carbodiimide hydrochloride (EDC-HCl, 98%) were purchased from Aladdin. *N*-Vinyl-2-pyrrolidone (NVP 98%) was obtained from J & K Scientific (USA). 1-Thioglycerol (TG, 99%) was obtained from TCI. Ethyl acetate hydrochloride (EA-HCl) was purchased from Adamas-Beta (Tokyo, Japan). Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2003). Marine bacteria (*Pseudomonas* sp.) (strain MCCC 1A00099) were purchased from the Maritime Culture Collection of China (MCCC), and the marine culture medium 2216E (HB0132-1) was purchased from Qingdao Hope Biotechnology Co., Ltd.

Synthesis of PDMS-based Polymers (P)

Poly(vinyl pyrrolidone) diol (PVPD), lysine-dopamine (LDA), *N*-vinyl-2-pyrrolidone (NVP)-dodecafluoroheptyl methacrylate (DFMA) diol (PVPFAD) were synthesized as previously described.^[40,43,44] The characterization data for PVPD, LDA, and PVPFAD are shown in Figs. S1–S4 (in the electronic supplementary information, ESI).

Typically, IPDI (0.854 mL, 4.08 mmol) was dissolved in THF (5 mL) under the protection of argon, and then the PVPD (0.902 g, 0.8 mmol) was added and reacted with IPDI at 80 °C for 9 h. LDA (0.4 mmol, 0.142 g) was dissolved in ultradry DMF (10 mL) and triethylamine was then added dropwise. The solution was aspirated using a syringe and added dropwise to the IPDI. The mixture was then placed in an ice water bath for 3 h. The PDMS was heated at 120 °C under vacuum for 2 h. Subsequently, PDMS (2.8 mmol, 11.136 mL) was transferred to a dropping funnel and ultra-dry THF (40 mL) was added. The cooled PDMS was then added dropwise to the mixed solution and stirred at 25 °C under Ar for 24 h. The product was labeled P4.

Preparation of Self-healing Antifouling Polymers (P/ Fe^{3+})

The PDMS-based polymer (P) is dissolved in a defined amount of THF solution, then FeCl_3 (0.01 mol·L⁻¹) solution was added dropwise to the polymer solution; the pH of the solution was

adjusted to 10 with triethylamine. The Fe^{3+} /LDA molar ratio is 1:5. The mixed solution was then poured into a mold and the solvent was evaporated at 25 °C for 24 h to obtain P/ Fe^{3+} . The samples were then placed in a vacuum oven and dried for 24 h. The desired coating was obtained by dissolving P/ Fe^{3+} (0.15 g) in THF (1.5 mL) and applying the solution to a Q235 carbon steel substrate (3 × 4 cm). The prepared samples were then dried at 25 °C for 24 h.

Characterization

¹H- and ¹⁹F nuclear magnetic resonance (¹⁹F-NMR) spectra were recorded on a Bruker AVANCE III spectrometer (600 MHz) using CDCl_3 and DMSO-d_6 as solvents with tetramethylsilane (TMS) as the internal standard. Fourier-transform infrared (FTIR) spectra were obtained from 4000 cm⁻¹ to 500 cm⁻¹ using a Thermo Scientific Nicolet 6700 spectrometer equipped with an attenuated total reflectance (ATR) detector. The molecular weight and PDI were determined by gel permeation chromatography (GPC) (Waters 1515) using THF as the mobile phase. UV-Vis absorption spectroscopy was performed using a Thermo Scientific Evolution 220 spectrophotometer (Thermo Scientific). Small-angle X-ray scattering (SAXS) measurements were conducted using an SAXS-Xuess2-0. The X-ray wavelength is 1.5418 Å. The water contact angle (WCA) was determined by measuring five different regions in each sample using an OCA50 Dataphysics instrument. The surface free energy was obtained using the Owens-Wendt-Rabel-Kaelble (OWRK) method. Diiodomethane and deionized water were used as the test liquids. The volume of the test liquid was 5 µL. The adhesion tests were performed using a pull-off adhesion tester (BGD-500).

The surface topography and elemental composition of the samples were analyzed using a JEOL JSM-7610F scanning electron microscope (SEM). Dumbbell-shaped samples were subjected to tensile tests at 80 mm/min using a ZwickRoell instrument (Z1.0). The healing efficiency (η) was calculated using Eq. (1).

$$\eta = (\sigma_{\text{heal}} / \sigma_{\text{ori}}) \times 100\% \quad (1)$$

where σ_{ori} and σ_{heal} are the tensile strengths of the original and the healed samples, respectively. The tests were conducted in triplicate, and the average values were used. Microscopic images were recorded using an optical microscope (Axioskop 40 A Pol, Nikon). Scratches were made on the coatings using a cutter blade.

The antifouling properties of the coatings were evaluated using marine bacteria (*Pseudomonas* sp.). The polymer tetrahydrofuran solution was uniformly coated onto a pre-prepared 1 cm × 1 cm glass substrate and dried for subsequent use. Before inoculation, the 2216E medium was autoclaved at 121 °C for 20 min (MLS-3781-PC, Panasonic). A single-edged blade was used to scratch the coated surface of a fiberglass board (5×0.02 mm), and the polymer-coated fiberglass panels were immersed in a suspension of marine bacteria (*Pseudomonas* sp.) (1 mL, 10⁸ cells/mL) at 30 °C for 24 h. The cells were then stained with SYTOTM 9 green fluorescent nucleic acid stain (Thermo Fisher Scientific). The dye (3 µL) was diluted 10⁵-fold with sterile water. Subsequently, the bacteria were incubated in staining solution for 40 min in the dark. Adherent bacterial cells were examined under a fluorescence microscope (Scope A1, Zeiss). The incubated *Pseudomonas* sp. was removed, diluted to 10⁵ cells/mL, immersed in the diluted bacterial solution, and incubated at a constant

temperature for 24 h. The sample was subsequently removed and the bacteria on the surface of the coating were dislodged using a vortex mixer. Then, 50 μL of the bacterial suspension was pipetted onto an agar plate. Adherent bacteria were examined using confocal laser scanning microscopy (TCS-SP8).

RESULTS AND DISCUSSION

Synthesis and Characterization of Self-healing Antifouling Polymers

N-Vinyl-2-pyrrolidone (NVP), 1-thioglycerol (TG), and dodecafluoroheptyl methacrylate (DFMA) are functional monomers used

for the synthesis of PVPD and NVP-DFMA diol (PVPFAD).^[45] PVPFAD was synthesized by changing the molar ratio of NVP to DMFA. Lysine-dopamine (LDA) was synthesized according to a previously reported procedure.^[43,44] The chemical structure was characterized using ^1H nuclear magnetic resonance (^1H -NMR), ^{19}F -NMR, and FTIR spectroscopic analyses (Figs. S1–S4 in ESI). PDMS, isophorone diisocyanate (IPDI), LDA, and either PVPD or PVPFAD were utilized for further polymerization reactions to prepare PDMS-based polymers (P) (Fig. 1 and Fig. S5 in ESI). Subsequently, the as-prepared polymers were pre-crosslinked with Fe^{3+} to form self-healing antifouling polymers denoted as P/ Fe^{3+} .

Table 1 and Table S1 (in ESI) list the compositions, molecu-

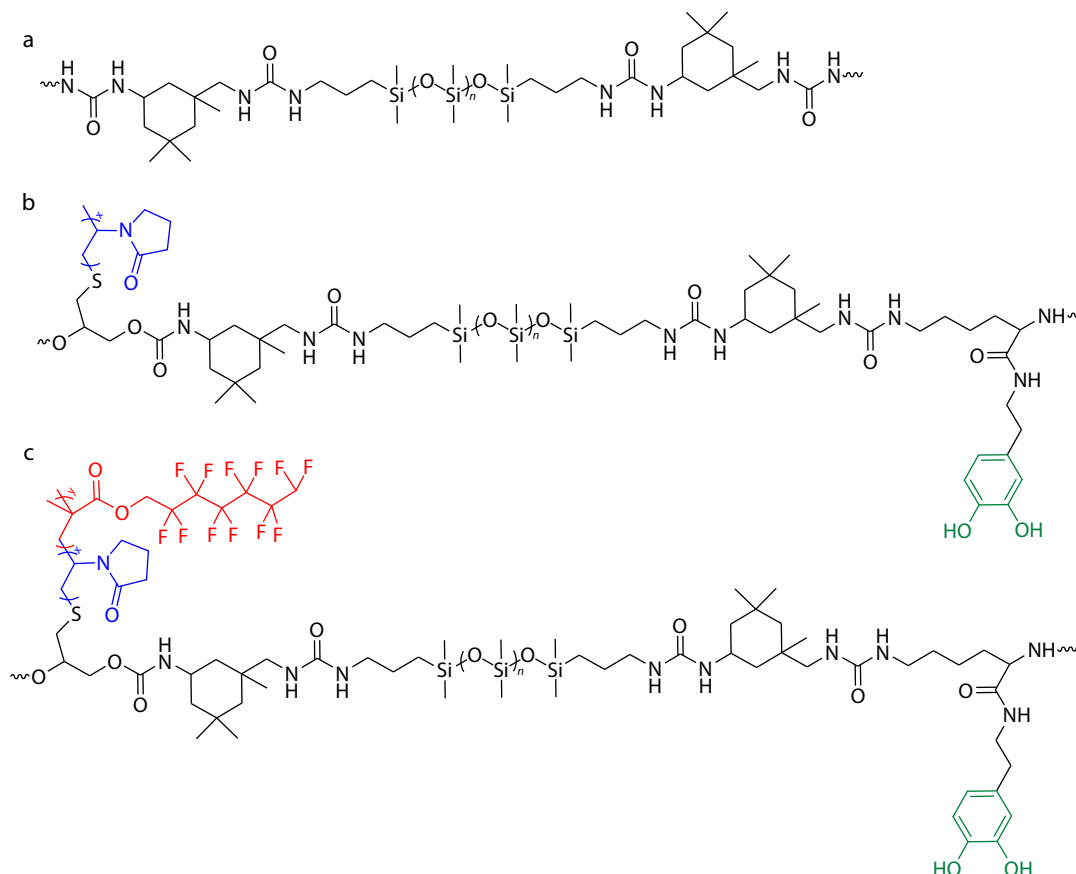


Fig. 1 Chemical structure of the polydimethylsiloxane (PDMS)-based polymers (P) for the following preparation of the self-healing antifouling materials: (a) P0, (b) P1–P8, (c) P9–P11.

Table 1 The chemical composition, molecular weight, and polydispersity index (PDI) of the synthesized polymers.

Sample	IPDI/PDMS/LDA/PVPD ^a	M_n^b (10^4 g·mol ⁻¹)	M_w^b (10^4 g·mol ⁻¹)	PDI ^b
P0	10.2/10/0/0	3.21	6.48	2.02
P1	10.2/9/1/0	4.22	7.58	1.80
P2	10.2/8.5/1/0.5	3.70	7.39	2.00
P3	10.2/8/1/1	2.71	5.19	1.92
P4	10.2/7/1/2	3.36	7.03	2.09
P5	10.2/8/0/2	2.43	4.62	1.90
P6	10.2/7.5/0.5/2	2.34	4.70	2.01
P7	10.2/6/2/2	2.82	7.06	2.50
P8	10.2/6/1/3	2.87	7.14	2.49

^a Molar ratio of isophorone diisocyanate (IPDI), polydimethylsiloxane (PDMS), lysine-dopamine (LDA), and poly(vinyl pyrrolidone) diol (PVPD); ^b Determined by gel permeation chromatography (GPC); The mobile phase was tetrahydrofuran (THF).

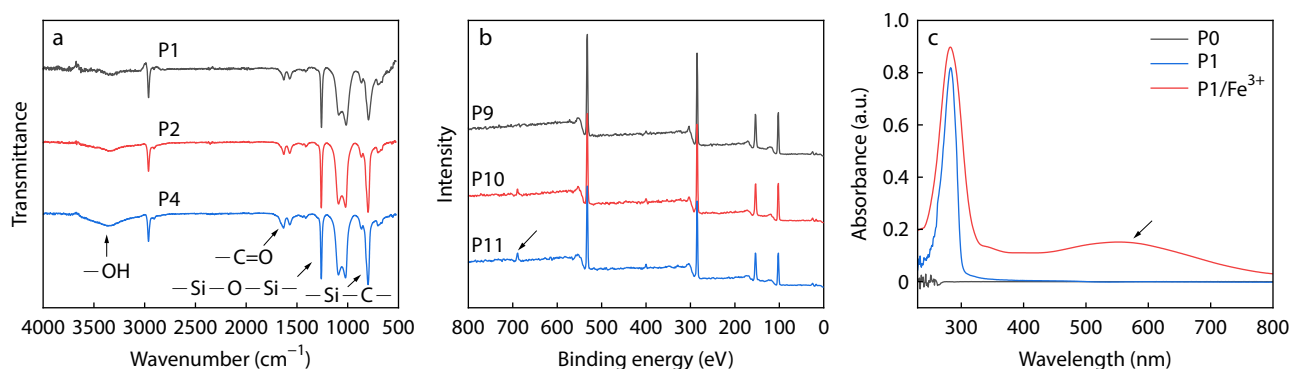


Fig. 2 Structural characterization of PDMS-based polymers: (a) Fourier-transform infrared (FTIR) spectra of P1, P2, and P4; (b) X-ray photoelectron spectroscopy (XPS) spectra of P9, P10, and P11; (c) Ultraviolet-visible (UV-Vis) absorbance spectra of P0, P1, and P1/Fe³⁺.

lar weights, and polydispersity indices (PDI) of the synthesized polymers. Their chemical compositions were adjusted by varying the feed ratios of PDMS, LDA, IPDI, and PVPD or PVPFAD. Their molecular weights range from 1.3×10^4 g·mol⁻¹ to 4.2×10^4 g·mol⁻¹, with a polydispersity index (PDI) ranging from 1.81 to 2.62.

The chemical structures of the synthesized polymers were further validated by FTIR spectroscopy, ultraviolet-visible (UV-Vis) spectroscopy, X-ray photoelectron spectroscopy (XPS), and Raman spectroscopy. The FTIR spectra shown in Fig. 2(a) reveal the absorption peak of -OH in water at 3300–3500 cm⁻¹ owing to the hygroscopic nature of the PVPD component.^[45] The stretching vibrations of -Si-C- (800 cm⁻¹), -C-F (1180 cm⁻¹) (Fig. S6 in ESI), and -Si-O-Si- (1260 cm⁻¹) were also detected. In the XPS spectra shown in Fig. 2(b), the F 1s peak appears at 685.7 eV, indicating the successful introduction of DMFA.^[46] The UV-Vis spectra shown in Fig. 2(c) reveal that no absorption peaks are observed with P0, whereas samples P1 and P1/Fe³⁺, containing the LDA unit, display a distinct absorption peak at 280 nm, characteristic of the catechol group.^[47,48] These results indicated that polymers containing catechol groups were synthesized. Thus, spectroscopic analyses confirmed the synthesis of the target polymers containing the LDA and PVPD or PVPFAD segments.

Moreover, characteristic peaks at 585 and 633 cm⁻¹, originating from the Fe-O bond vibrations of Fe³⁺-catechol interactions,^[49] were observed in the Raman spectra of P4/Fe³⁺ (Fig. S7 in ESI). After incorporating Fe³⁺, an absorption band emerged between 500 and 700 nm in the UV-Vis spectra (Fig. 2c), indicating the formation of a bis-/tris-complex of Fe³⁺-catechol.^[50] These results confirm that PDMS-based polymers with Fe³⁺-catechol cross-linking points were successfully prepared.

Morphology and Antifouling Properties of the Self-healing Antifouling Polymers

The surface morphologies of the samples were investigated by atomic force microscopy (AFM), small-angle X-ray scattering (SAXS), and energy-dispersive spectrometry (EDS). As shown in Fig. S8 (in ESI), P10 (containing PVPFAD fragments) was observed by EDS to form nanodomains and disperse them in the PDMS matrix. Fig. 3 shows the SAXS profiles of P1, P4, and P10. No scattering signals are observed for P1, whereas scattering peaks are observed for P4 and P10. These results demonstrated that a microphase-separated morphology was formed after the

incorporation of hydrophilic PVPD or fluorinated amphiphilic PVPFAD into the PDMS matrix. It has been reported that surface compositional heterogeneities and microphase-separated domains on the surface significantly influence the protein adsorption behavior, thereby conferring antifouling properties to the coatings.^[51,52]

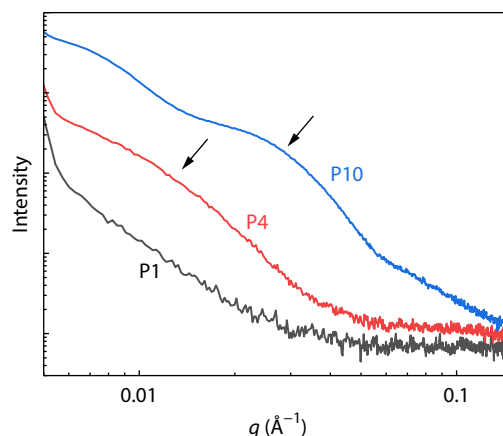


Fig. 3 Microphase-separated structure of self-healing antifouling polymers. Small-angle X-ray scattering (SAXS) spectra of P1, P4, and P10.

Fig. 4 shows the surface and antifouling properties of the coatings. As shown in Fig. 4(a), as the PVPD content increases, the water contact angle decreases from 104.1° to 81.1°. In addition, as shown in Fig. 4(b), the adhesion of the coating gradually increases with increasing fluorine content in the PVPFAD; the adhesion reaches a maximum of approximately 3.1 MPa with NVP to DMFA feed molar ratios of 4:4 and 4:8. The surface energy of the polymer also varied with relative NVPD content (Fig. S9 in ESI). Fig. 4(c) shows that the number of attached bacterial colonies on the coating decreased with an increase in the amount of PVPD. The colony density of samples P1/Fe³⁺ to P4/Fe³⁺ plates decreased from 9.7×10^7 cells/mL to 1.6×10^7 cells/mL. This suggests that the as-prepared self-healing antifouling polymer coating exhibits efficient antibacterial performance.

Mechanical and Self-healing Properties of the Materials

The original mechanical properties of the P/Fe³⁺ samples were examined using tensile testing. As depicted in Figs. 5(a), 5(b),

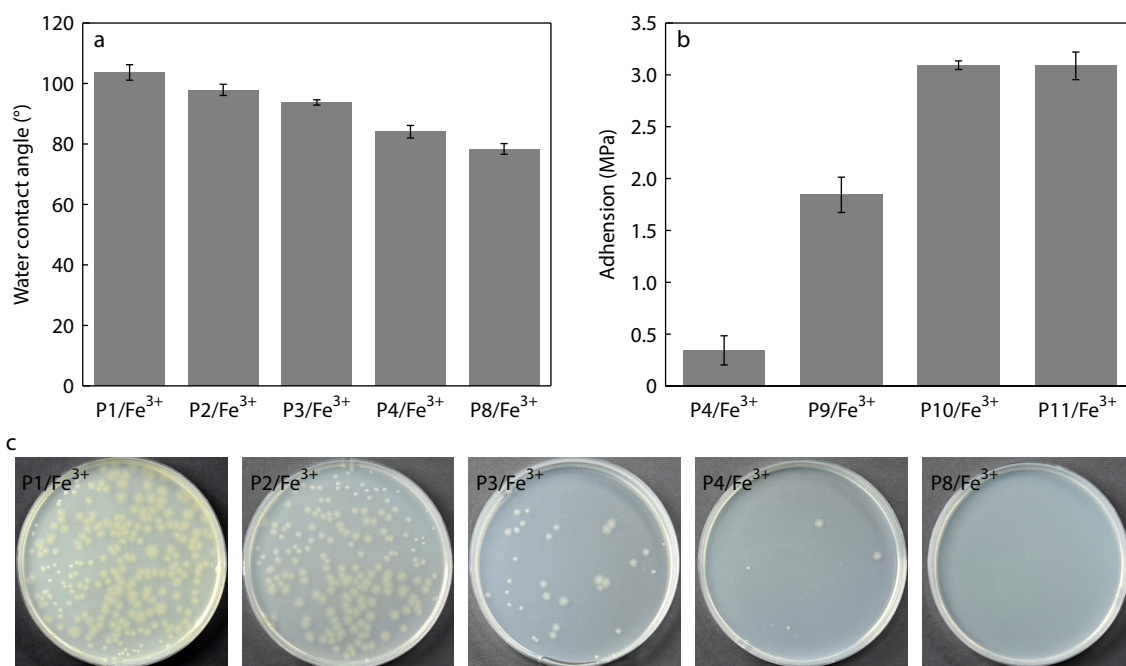


Fig. 4 Surface, adhesion, and antifouling properties of the self-healing antifouling materials: (a) contact angles of self-healing antifouling materials after 7 days of immersion in artificial seawater (ASW); (b) adhesion strengths of samples P4/Fe³⁺, P9/Fe³⁺, P10/Fe³⁺ and P11/Fe³⁺; (c) colonies of *Pseudomonas* sp. formed on agar plates with different coatings.

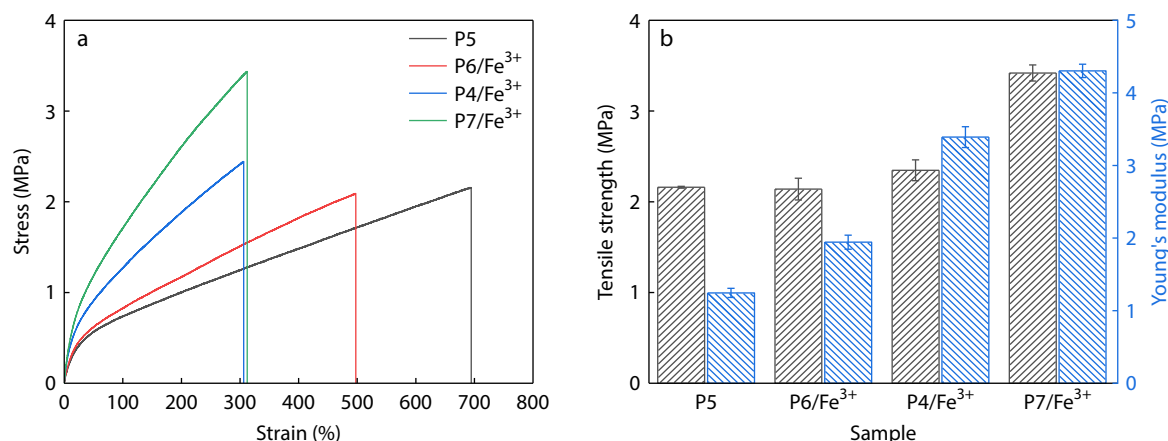


Fig. 5 Mechanical properties of the materials: (a) stress-strain curves and (b) corresponding tensile strength and Young's modulus of P5, P6/Fe³⁺, P4/Fe³⁺, and P7/Fe³⁺.

S10 (in ESI), and Table S2 (in ESI), the tensile strength and Young's modulus of the polymers increased from 2.16 MPa to 3.42 MPa and from 1.24 MPa to 4.30 MPa, respectively, as the amount of Fe³⁺-catechol crosslinking increased. This enhancement was attributed to both the introduction of a rigid benzene structure in catechol and the increase in the number of Fe³⁺-catechol crosslinking points in the material network.

To further assess the self-healing properties of the samples, morphological changes upon scratching and tensile tests were conducted after self-healing. The coatings were prepared on glass plates, scratched, and immersed in artificial seawater (ASW) for different durations. As illustrated in Figs. 6(a) and S11 (in ESI), the scratches on P4/Fe³⁺ almost completely disappeared after immersion in ASW for 24 h. In addition, when a dumbbell-shaped sample was cut, reattached,

and immersed in ASW for 24 h, the repaired sample regained its stretchability and could be extended to 300% of its original length (Fig. 6b). As shown in the stress-strain curves of the original and repaired P4/Fe³⁺ (Fig. 6c), the profile returned to its original state after 24 h.

Self-healing efficiencies were calculated to evaluate the self-healing performance quantitatively. The results are shown in Figs. 6(d) and S12 (in ESI). As the immersion time in ASW increased, the self-healing efficiency gradually increased. In addition, with an increase in the number of dynamic Fe³⁺-catechol crosslinks, the self-healing efficiencies first increased and then decreased. For P4/Fe³⁺, the self-healing efficiency reached approximately (98±11)% after self-healing for 24 h. This self-healing behavior originates from the dynamic reassociation of Fe³⁺-catechol crosslinks under

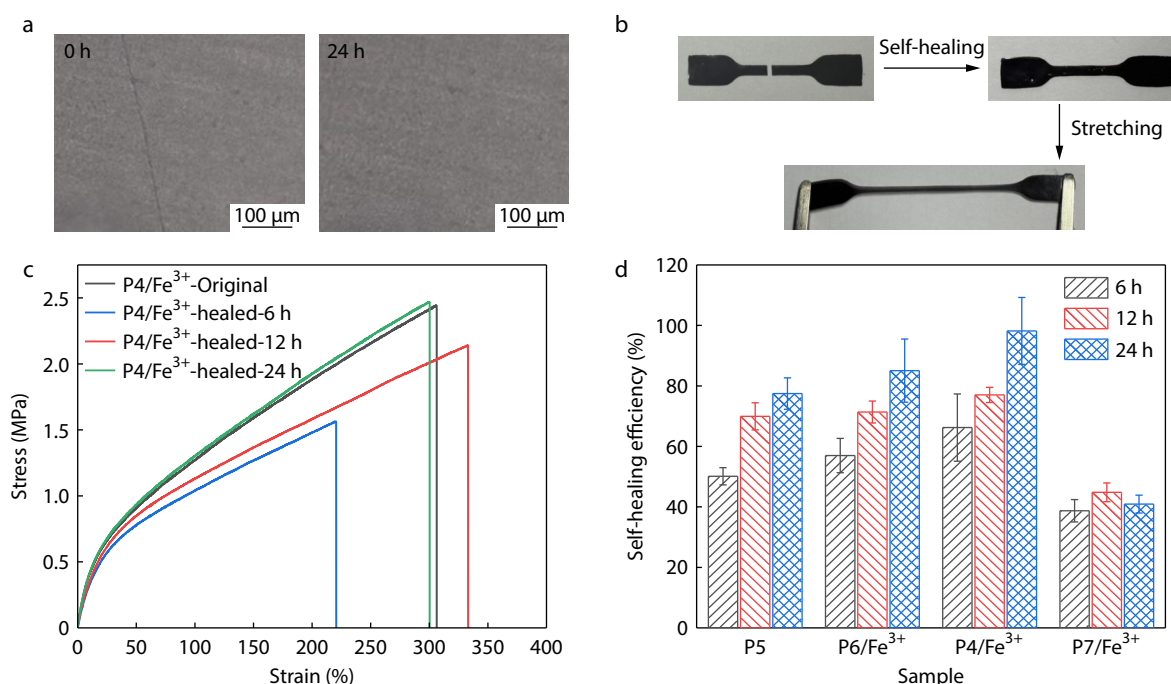


Fig. 6 Self-healing properties of the as-prepared materials: (a) optical microscopic images of P4/Fe³⁺ in artificial seawater (ASW) for 24 h; (b) images of damaged, self-healing, and stretched P4/Fe³⁺ dumbbell-shaped samples. The sample was stretched to 300% of its original length after self-healing in ASW for 24 h; (c) stress-strain curves of P4/Fe³⁺ after self-healing for different durations; (d) effect of immersion time in ASW on self-healing efficiencies of P5, P6/Fe³⁺, P4/Fe³⁺, and P7/Fe³⁺.

water. For samples with different PVPD contents, the analysis revealed that P1–P3/Fe³⁺ exhibited self-healing efficiencies higher than 80% after 6 h (Fig. S12b in ESI). For P9/Fe³⁺ and P10/Fe³⁺ with different fluoroalkyl contents, the self-healing efficiency exceeded 80% after immersion for 24 h (Fig. S12c in ESI). This demonstrates that the incorporation of hydrophobic fluoroalkyl and hydrophilic PVPD segments had a minimal influence on the self-healing properties.

Self-healing Antifouling Properties of the Materials

The marine bacterium *Pseudomonas* sp. was used for the antibacterial experiments. The results are presented in Figs. 7, S13, and S14 (in ESI). Uniform scratches were introduced to the coating, and the self-healing antifouling properties were assessed by evaluating bacterial attachment in the damaged areas. As illustrated in Fig. S13 (in ESI), distinct scratches were evident on the surface, with *Pseudomonas* sp. adhering to both the surface and

the scratched areas of sample P0. For P5, which lacks Fe³⁺-catechol crosslinks, the scratch did not heal, resulting in the enrichment of numerous colonies at the site (Fig. 7a). In contrast, the surface scratches of P4/Fe³⁺ and P10/Fe³⁺ were effectively healed through the reconstruction of the dynamic crosslinks, and almost no bacteria adhered to the surface (Fig. 7b and S14 in ESI).

Based on the above results, a self-healing antifouling mechanism for the material was proposed, as illustrated in Fig. 8. When conventional materials are damaged, the affected areas lose their antifouling capabilities, leading to biofouling. Aggregation of biofouling may result in severe microbial corrosion of metal substrates. In contrast, when self-healing antifouling materials are damaged in aqueous environments, the Fe³⁺ and catechol groups at the damaged interface reassemble via coordination interactions, forming a cross-linked

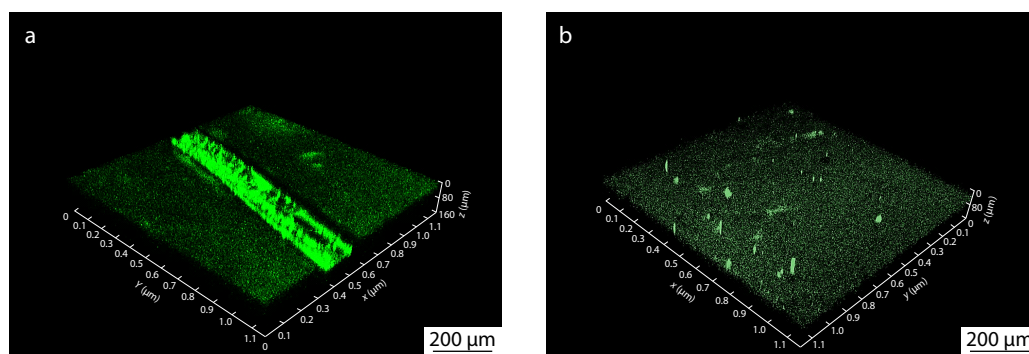


Fig. 7 Self-healing antifouling performance. Confocal laser scanning microscopy images of *Pseudomonas* sp. attachment on coating surfaces of (a) P5 and (b) P4/Fe³⁺.

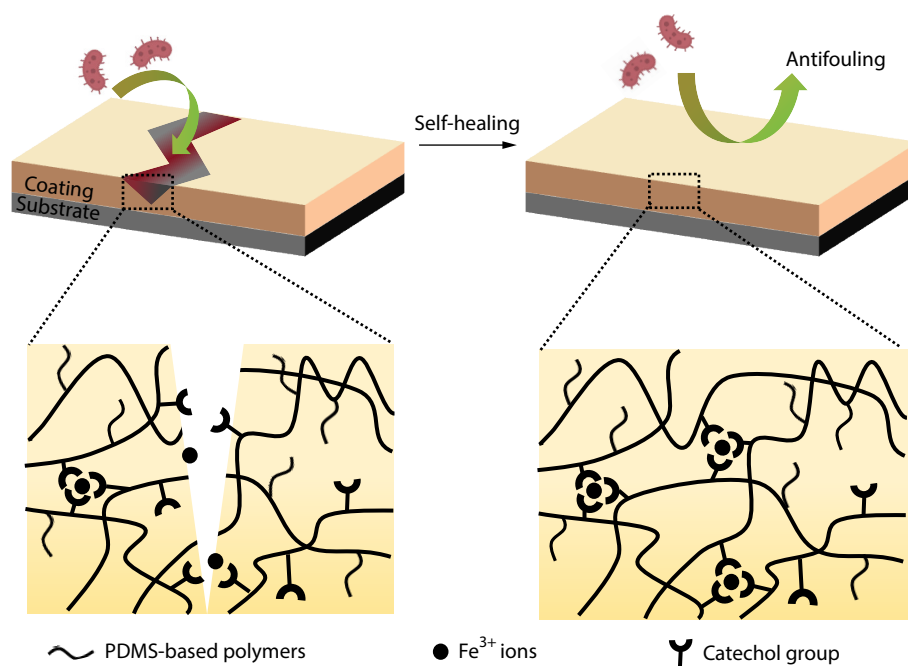


Fig. 8 Schematic diagram of the self-healing antifouling mechanism.

network that repairs the damage. Once the damage is repaired, the reformation of a microphase-separated structure on the surface restores the antifouling properties and prevents adhesion of biological fouling.

CONCLUSIONS

PDMS-based antifouling polymers with tunable self-healing properties in an aqueous environment were constructed by introducing hydrophilic PVPD segments or fluorinated amphiphilic segments and Fe^{3+} -catechol dynamic coordination crosslinks. The microphase-separated morphology within the PDMS matrix endowed the coating with a repellent of the marine bacteria *Pseudomonas* sp. The Fe^{3+} -catechol coordination interaction enabled the restoration of its mechanical properties with a self-healing efficiency of nearly 100% after 24 h in aqueous environments at room temperature. In contrast to the accumulation of bacteria in the damaged region of the control sample, almost no bacteria were observed in the repaired region of the self-healing antifouling coating. The long-term stability of the antifouling property and fatigue resistance of the dynamic bonds will be further evaluated to ensure their practical applicability. Our strategy provides valuable insights into the development of innovative self-healing antifouling coatings and a promising outlook for marine vessels and equipment.

Conflict of Interests

The authors declare no interest conflict.

Electronic Supplementary Information

Electronic supplementary information (ESI) is available free of charge in the online version of this article at <http://doi.org/>

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Data Availability Statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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